Phan Huy Hoang | Electronics Engineer

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Summary

Passionate electronics engineer with expertise in embedded systems, IoT, and electronic circuit design. Strong analytical and problem-solving skills, with hands-on experience in hardware development, firmware programming, and smart devices integrated with AI. Always eager to learn emerging technologies and contribute to innovative, real-world solutions.

Education

Ho Chi Minh City University of Technology and Education
Major in Software Engineering *2021 - Present*
English Certificate B1

Work Experience

Daklak Industrial Zone
Maintenance Engineer *06/2024 - 08/2025*
- Proficient in working with Programmable Logic Controllers (PLC) and well-versed in high-power electrical systems and industrial automation.

Semicon Labs
Internship *05/2025 - 08/2025*
- Designed, developed, and maintained corporate websites using WordPress, improving UI/UX design.
- Performed SEO optimization, boosting search rankings and increasing site traffic by 25%.

Ampere Technologies
Internship *06/2023 - Present*
- Developed and optimized high-efficiency power systems for various industrial applications.
- Worked on microcontroller-based solutions, focusing on energy-efficient designs and automation.

Personal Projects

Simulate a traffic light system on an FPGA board:
- Designed and implemented a traffic light system using Verilog for FPGA, simulating a cycle of red, green, and yellow lights.
- Developed a state machine to control light transitions based on preset timing intervals.
- Used clock-driven counters to track the timing of each light (red for 3 seconds, green for 3 seconds, yellow for 1 second).
- Technologies: Verilog, FPGA, State Machine, Timing Control.

PWM Modulation and Communication Systems:
- Designed and implemented a PWM modulation system for controlling motors in embedded systems.
- Developed communication protocols using SPI and I2C for sensor and actuator integration.
- Created and optimized digital filters for signal processing in embedded systems.
- Technologies: C++, Arduino, Raspberry Pi.

High-Speed Communication System with SerDes:
- Developed a high-speed communication system using SerDes (Serializer/Deserializer) technology for data transfer.
- Integrated an ARM-based SoC with FPGA to implement complex signal processing algorithms.
- Optimized hardware and firmware for reliable data transmission at high speeds.
- Technologies: Verilog, C, ARM, FPGA.

IoT Fire Alarm System:
- Designed an IoT-based fire alarm system with real-time monitoring and notifications.
- Integrated sensors, including smoke and gas detectors, with cloud-based data storage for analysis.
- Technologies: Arduino, IoT, MQTT, Cloud Computing.

PLL Optimization in Electronics Circuits:
- Optimized phase-locked loop (PLL) circuits to improve frequency stability in analog electronics.
- Implemented advanced techniques for reducing jitter and enhancing signal integrity.
- Technologies: Verilog, Analog Circuit Design.

Skills

Programming: C,Verilog, System Verilog, Shell Bash *Proficient*
Development Tools: Xilinx, ModelSim, QuasterSim, Linux *Proficient*
Soft Skills: Communication, Presentation, Teamwork *Strong*